

Jan Kwong

jankwong705@gmail.com | +1 (509) 968-8888 | <https://jankwong705.github.io/me>

EDUCATION

University of California, San Diego

M.S. in Computer Science (Anticipated). GPA: 3.8.

La Jolla, CA

Dec 2026

University of California, San Diego

B.S. in Computer Science. GPA: 3.8.

La Jolla, CA

Jun 2025

PROJECTS

picoLM: Compact Language Model

Apr 2026 – Jun 2026

- Pretrained a 98M-parameter decoder-only transformer (GQA, SwiGLU, RoPE) in PyTorch on a single shared, interruptible GPU, achieving ~41 perplexity on a 5M-token web-text validation set
- Built a streaming data pipeline over FineWeb, SlimPajama, DCLM, and RedPajama-V2 with on-the-fly hash deduplication and token packing, never materializing the corpus on disk
- Applied knowledge distillation from GPT-Neo-2.7B, reducing perplexity from 70 to 42 by combining teacher-imitation and ground-truth losses

RAG & Context Engineering Pipeline

Apr 2026 – May 2026

- Built a single-turn Q&A Retrieval-Augmented Generation system over a technical OSS documentation corpus, enforcing a strict 2,000-token per-query context budget
- Conducted a systematic sweep of 8+ configurations (varying chunk size, embedding model, retrieval strategy, and reranker settings) to optimize retrieval and generation quality
- Achieved 0.93 Correctness and 1.00 Faithfulness by implementing parent-document retrieval with BM25/vector hybrid search, a BGE cross-encoder reranker, and a dual-citation few-shot prompting strategy

Security Analysis of Apple Intelligence

Oct 2025 – Dec 2025

- Reverse-engineered macOS binaries using Ghidra to map information flow between user input, generative AI models, and local storage
- Identified a privacy vulnerability where the system persisted unencrypted sensitive personal information to local directories, bypassing the ephemeral-only privacy policy
- Developed a Python forensic script to simulate TCC (Transparency, Consent, and Control) inheritance attacks, successfully exfiltrating sensitive user data from sandboxed environments

EXPERIENCE

Graduate Teaching Assistant

Apr 2026 – Jun 2026

UC San Diego

La Jolla, CA

- Led weekly discussion sections of ~40 students for the class Intro to Research Methods, mentoring them through empirical study design, statistical validation, and analysis methodologies
- Designed and refined discussion slides covering key course concepts, hosted weekly office hours, and gave students concrete feedback on assessments, helping them grasp difficult ideas and improve their work

Computer Science Tutor

Apr 2024 – Jun 2025

UC San Diego

La Jolla, CA

- Supported ~200 students in Theory of Computation and Advanced Data Structures through office hours, clarifying difficult concepts
- Evaluated homework and exams with detailed feedback, and met weekly with a 10-person teaching team to address course progress and challenges

Undergraduate Research Assistant

Jan 2024 – Dec 2024

UC San Diego LASR Lab

La Jolla, CA

- Co-authored a paper (submitted to Acoustical Society of America and PsyArXiv, under review) benchmarking OpenAI Whisper against 75 human transcribers on 300 English sentences from 20 diverse accented speakers
- Automated WER computation with Python scripts, discovering that humans significantly outperform models on isolated words, suggesting potential limitations in models' training data or acoustic context requirements

SKILLS

ML & AI: PyTorch, Hugging Face, Scikit-learn, Machine Learning, NLP, Computer Vision, Data Engineering, Data Analytics, LLM Pretraining, Knowledge Distillation, RAG

Programming Languages: Python, SQL, Java, C, C++, JavaScript, HTML, CSS, ARM, Clojure, Rocq

Tools & Infrastructure: Pandas, NumPy, Git, Docker, Kubernetes, Ghidra, CI/CD

CS Foundations: Data Structures, Algorithms, Database Systems, Operating Systems, Computer Architecture, Computer Security, Agile

Languages: Cantonese (Native), English (Native), Mandarin (Native), Japanese (Advanced)